weekly consumption and cost for a home dashboard. Rather than running aggregation queries every time the dashboard loads, I can use Apache Spark to transform and aggregate the raw CSV data once, as shown in listing 4.7. Apache Spark's distributed processing capabilities ensure this approach scales to very large data sets.

Listing 4.7 An example of using Apache Spark to extract minutely power consumption readings, convert their units, and aggregate to weekly totals.

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import date_trunc, sum as spark_sum, avg, col
from pyspark.sql.types import *

spark = SparkSession.builder.appName("WeeklyUsage").getOrCreate()

schema = StructType([ #1
    StructField("datetime_utc", TimestampType(), True),
    StructField("datetime_local", TimestampType(), True),
    StructField("watt_hours", DoubleType(), True),
    StructField("cost_dollars", DoubleType(), True),
    StructField("is_peak", BooleanType(), True)
])

df = spark.read.option("header", "true").schema(schema).csv(
  ➥"powerpal_data.csv")

weekly_usage = df.groupBy( #2
    date_trunc("week", col("datetime_local")).alias("week_start")
).agg(
    (spark_sum("watt_hours") / 1000).alias("total_kwh"),
    spark_sum("cost_dollars").alias("total_cost_dollars")
).orderBy("week_start")

print("Weekly Usage (kWh) and Cost:")
weekly_usage.show()

spark.stop()
```

#1 Define the structure of the data stored in the CSV file.
#2 Spark's SQL functions group and aggregate the data in the CSV file.

Modern *ELT* approaches load raw data first and perform transformations within the target system. This pattern works well with cloud data warehouses that provide powerful transformation capabilities and can handle raw data volumes efficiently.

4.5 Storage solutions

Selecting the right storage solution is a key factor in the success of your digital twin meeting your objectives. Digital twins require diverse data types, from real-time sensor readings and historical logs to design blueprints and simulation outputs, each with different storage requirements for capacity, performance, scalability, and security.